OryzaFlow – System Understanding

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What OryzaFlow is

OryzaFlow is a digital platform for Sri Lanka's paddy (நெல்) supply chain, from the field to the mill. It is based on two source documents: the Business Description & Business Model (7 pages) and the Project Proposal (9 pages).

Today the trade runs on phone calls, trust and paper. That causes lorry queues at mill gates, arguments over weight and moisture, late payments, and trucks and labour sitting idle. OryzaFlow replaces this with one flow: **book → measure → track → receive → pay**, with a neutral, time-stamped record at every step.

Core trip flow

Every delivery follows the same seven steps, and payment is released only when the mill confirms receipt.

1. **The mill posts demand and price.** The dealer (muthali/retailer) sees it in real time.
2. **The dealer books a slot at the mill**, plus a lorry and loading crew, in one request.
3. **IoT measurement at loading.** Weight, moisture and quality readings are saved automatically to the booking.
4. **The dealer checks the readings and confirms.** The booking goes to the mill and the driver.
5. **The bank blocks the payment** in the mill's account. The bank holds the money, not OryzaFlow.
6. **Live GPS tracking.** This covers the route, ETA, and driver behaviour (speed, harsh braking, going off route).
7. **Receiving at the mill.** The paddy is unloaded in the booked slot, re-weighed and checked, and the mill worker confirms. **The payment is then released** to the dealer straight away.

The 4 faces + IoT

Each of the planned faces maps to a role already described in the documents.

Face	In the docs	Main jobs
PC dashboard – mill owner	Mill intake portal + analytics (M5, M8)	Post demand and price, manage slots, see incoming lorries and ETAs, intake records, compare dispatch vs received readings, approve payments, season reports
Mobile – sending dealers	Muthali / retailer	Post loads, book slot and lorry, view and confirm IoT readings, track the lorry, get payment status
Mobile – transport drivers	Truck owners / drivers	Accept or reject jobs, navigate, GPS tracking, upload photo proof, update status
Mobile – receiving mill workers	Mill admin officer / unloading labour	Gate queue, check in arriving lorries, take receiving IoT readings, confirm receipt (which releases payment)
IoT	IoT layer	Weight (kg), moisture %, quality, GPS, driver behaviour, all tied to the booking ID

Everyone gets a unique user code after KYC (identity checks). The code sets their role and where they sit, for example workers under their mill and drivers under their lorry owner.

Observations

- **The two documents disagree a little.** The Business Model has the bank block and release the money. The Proposal has a simpler digital ledger with wallet and bank-transfer links. One must be picked for the first version.
- **Brokers and loading labour crews** are in the docs but not in the 4 faces. Should they be left out, or added later?
- **Tech stack in the Proposal:** Flutter (mobile), React + TypeScript (web), NestJS + PostgreSQL/PostGIS + Redis (backend), with Google Maps, an SMS gateway and Firebase push. Apps should work in Tamil, Sinhala and English, and offline where the signal is weak.
- **Measurements need to be taken twice**, once at dispatch and once at receiving. Comparing the two is what settles disputes.

Open questions

- [ ] **"Inunity" of the paddy:** is this humidity, i.e. moisture content %? What other readings are needed (foreign matter, broken grains, grain temperature, bag count)?
- [ ] **IoT placement:** at the dealer's loading point, on the lorry, at the mill gate, or more than one?
- [ ] **Hardware:** build our own (e.g. ESP32 + load cell + moisture sensor) or use ready-made scales and meters?
- [ ] **Stack:** stick with Flutter / React / NestJS / PostgreSQL from the Proposal?
- [ ] **Payments:** bank-blocked escrow or a simpler digital ledger for the first version?
- [ ] **Scope:** include brokers and loading crews now, or later?